

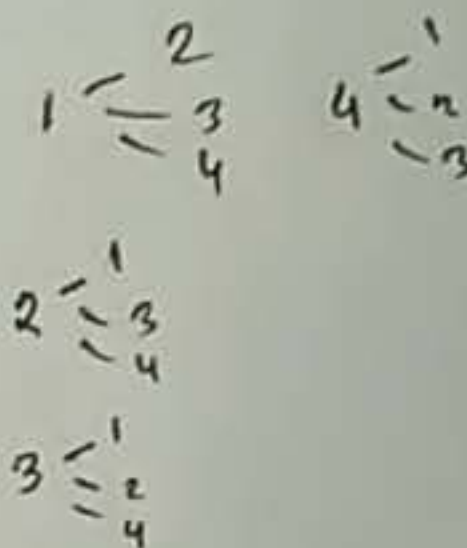
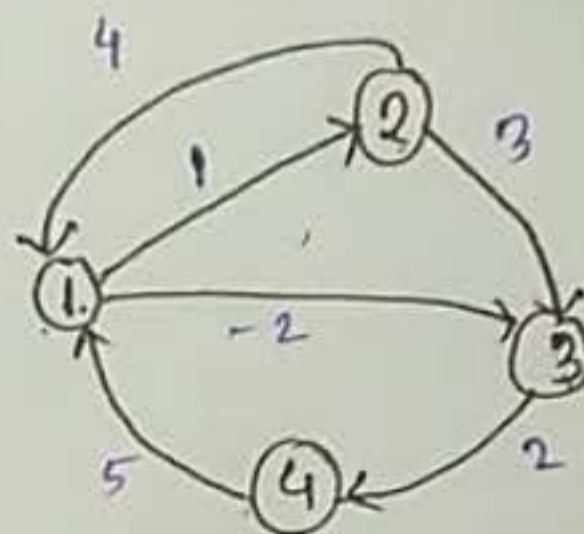
⇒ ① Works with negative & positive edges
(but with no negative cycle)

Made with KINEMASTER

Formula

$$D^k[i, j] = \min \{ D^{k-1}[i, j], D^{k-1}[i, k] + D^{k-1}[k, j] \}$$

We need to find matrix for each of the nodes and a distance matrix at the beginning. Matrix for the first node depends on the distance matrix, and each node's matrix depends on the previous node's matrix (as per the formula)

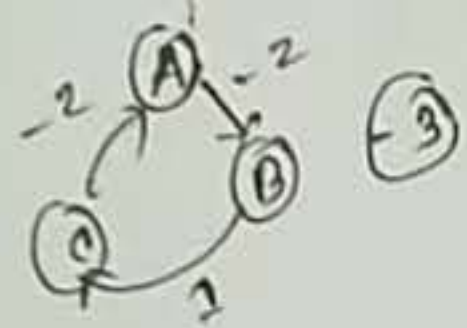


	1	2	3	4
1	0	1	-2	∞
2	4	0	3	∞
3	∞	∞	0	2
4	5	∞	∞	0

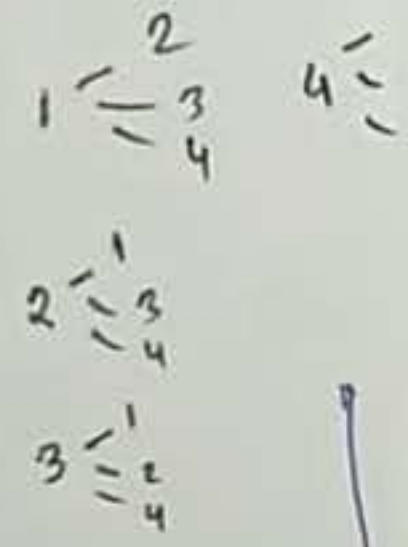
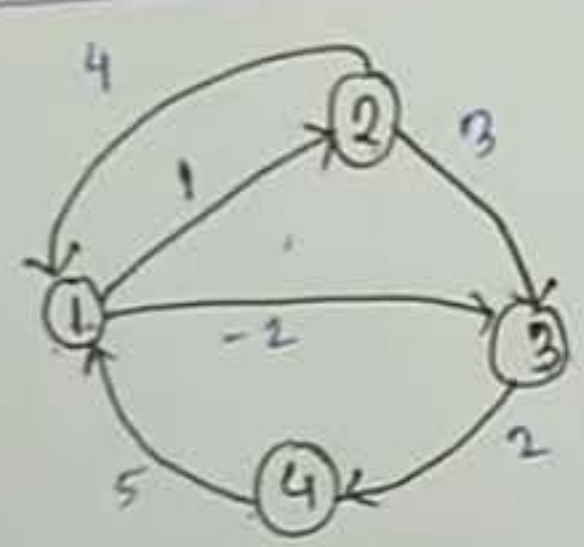
	1	2	3	4
1				
2				
3				
4				

...tive edges
cycle]

$$D^{k-1}[i, k] + D^{k-1}[k, j]$$



Floyd-Warshall
Made with **KINEMASTER**
All pairs shortest path



D²

	1	2	3	4
1				
2				
3				
4				

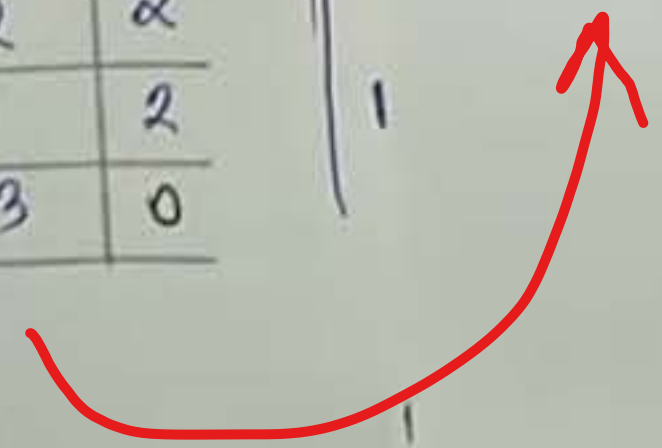
D³

	1	2	3	4
1				
2				
3				
4				

D¹

	1	2	3	4
1	0	1	-2	∞
2	4	0	2	∞
3	∞	∞	0	2
4	5	6	3	0

1



D^2

	1	2	3	4
1	0	1	-2	∞
2	4	0	2	∞
3	∞	∞	0	2
4	5	6	3	0

$$D^2[1,3] = \min\{D^1[1,3], D^1[1,2] + D^1[2,3]\}$$

$$= \min\{-2, 1 + 2 = 3\}$$

$$D^2[1,4] = \min\{D^1[1,4], D^1[1,2] + D^1[2,4]\}$$

$$= \min\{\infty, 1 + \infty = \infty\}$$

$$D^2[3,1] = \min\{D^1[3,1], D^1[3,2] + D^1[2,1]\}$$

$$= \min\{\infty, \infty + 4 = \infty\}$$

$$D^2[3,4] = \min\{D^1[3,4], D^1[3,2] + D^1[2,4]\}$$

$$= \min\{2, \infty + \infty = \infty\}$$

 D^3

	1	2	3	4
1	0	1	-2	
2		0	2	
3	∞	∞	0	2
4			3	0

$$D^3[1,2] = \min\{D^2[1,2], D^2[1,3] + D^2[3,2]\}$$

$$= \min\{1, -2 + \infty = \infty\}$$

$$D^3[1,4] = \min\{D^2[1,4], D^2[1,3] + D^2[3,4]\}$$

$$= \min\{\infty, -2 + 2 = 0\}$$

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 D^4

	1	2	3	4
1	0	1	-2	0
2		0	2	
3	∞	∞	0	2
4			3	0

All pairs shortest

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Made with KINEMASTER

D³

D⁴

1	2	3	4
0	1	-2	∞
4	0	2	∞
∞	∞	0	2
5	6	3	0

	1	2	3	4
1	0	1	-2	0
2	4	0	2	4
3	∞	∞	0	2
4	5	6	3	0

	1	2	3	4
1	0			∞
2		0		∞
3			0	2
4	5	6	3	0

$$D^3[1,2] = \min\{D^2[1,2], D^2[1,3] + D^2[3,2]\}$$

$$= \min\{-2, 1 + 2 = 3\}$$

$$D^3[1,4] = \min\{D^2[1,4], D^2[1,2] + D^2[2,4]\}$$

$$= \min\{\infty, 1 + \infty = \infty\}$$

$$D^3[3,1] = \min\{D^2[3,1], D^2[3,2] + D^2[2,1]\}$$

$$= \min\{\infty, \infty + 4 = \infty\}$$

$$D^3[3,4] = \min\{D^2[3,4], D^2[3,2] + D^2[2,4]\}$$

$$D^3[1,2] = \min\{D^2[1,2], D^2[1,3] + D^2[3,2]\}$$

$$= \min\{1, -2 + \infty = \infty\}$$

$$D^3[1,4] = \min\{D^2[1,4], D^2[1,3] + D^2[3,4]\}$$

$$= \min\{\infty, -2 + 2 = 0\}$$

$$D^3[2,1] = \min\{D^2[2,1], D^2[2,3] + D^2[3,1]\}$$

$$= \min\{4, 2 + \infty = \infty\}$$

$$D^3[2,4] = \min\{D^2[2,4], D^2[2,3] + D^2[3,4]\}$$

$$= \min\{4, 2 + 2 = 4\}$$



D³

	2	3	4
1		-2	∞
2	0	2	∞
3	∞	0	2
4	6	3	0

$$\min\{D^1[1,3], D^1[1,2] + D^1[2,3]\}$$

$$\min\{-2, 1 + 2 = 3\}$$

$$\min\{D^1[1,4], D^1[1,2] + D^1[2,4]\}$$

$$\min\{\infty, 1 + \infty = \infty\}$$

$$\min\{D^1[3,1], D^1[3,2] + D^1[2,1]\}$$

$$\min\{\infty, \infty + 4 = \infty\}$$

$$\min\{D^1[3,4], D^1[3,2] + D^1[2,4]\}$$

$$\min\{2, \infty + \infty = \infty\}$$

	1	2	3	4
1	0	1	-2	0
2	4	0	2	4
3	∞	∞	0	2
4	5	6	3	0

$$D^3[1,2] = \min\{D^2[1,2], D^2[1,3] + D^2[3,2]\}$$

$$= \min\{1, -2 + \infty = \infty\}$$

$$D^3[1,4] = \min\{D^2[1,4], D^2[1,3] + D^2[3,4]\}$$

$$= \min\{\infty, -2 + 2 = 0\}$$

$$D^3[2,1] = \min\{D^2[2,1], D^2[2,3] + D^2[3,1]\}$$

$$= \min\{4, 2 + \infty = \infty\}$$

$$D^3[2,4] = \min\{D^2[2,4], D^2[2,3] + D^2[3,4]\}$$

$$= \min\{\infty, 2 + 2 = 4\}$$

D⁴

Shortest path for all nodes

	1	2	3	4
1	0	1	-2	0
2	4	0	2	∞
3	7	8	0	2
4	5	6	3	0

$$D^4[1,2] = \min\{D^3[1,2], D^3[1,4] + D^3[4,2]\}$$

$$= \min\{1, 0 + 1 = 1\}$$

$$D^4[1,3] = \min\{D^3[1,3], D^3[1,4] + D^3[4,3]\}$$

$$= \min\{-2, 0 + 3 = 3\}$$

$$D^4[2,1] = \min\{D^3[2,1], D^3[2,4] + D^3[4,1]\}$$

$$= \min\{4, 4 + 5 = 9\}$$

$$D^4[2,3] = \min\{D^3[2,3], D^3[2,4] + D^3[4,3]\}$$

$$= \min\{\infty, 4 + 3 = 7\}$$